

# RhythmX Intelligence

## Detailed Architecture Design

A production-oriented design for authenticated wellness analytics, trusted trend retrieval, and constrained AI explanations.

| Architecture principle | Decision                                                                                           |
|------------------------|----------------------------------------------------------------------------------------------------|
| Facts before language  | PostgreSQL/Timescale queries and Python calculate values; Ollama only explains the compact result. |
| Existing data model    | Read directly from Django model-output tables. No new snapshot table is introduced.                |
| User isolation         | A gateway-authenticated numeric auth.User id scopes every health-data query.                       |
| Safety boundary        | Responses are non-medical, disclose incomplete data, and never expose database internals.          |

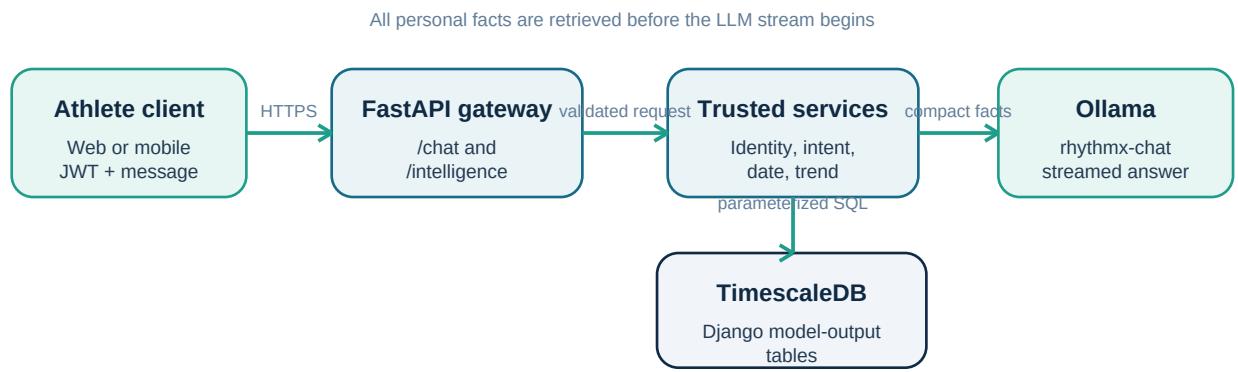
### Scope

The design covers historical recovery, readiness, and injury-risk questions through the existing FastAPI service. Current-day engine endpoints remain separate and continue to call their dedicated Ollama models.

Design baseline: September 2026

# 1. System Context

The gateway is the single orchestration boundary. It validates the caller identity, resolves intent and dates, obtains trusted numerical history, then supplies only the resulting analytics payload to the chat model.



| Component       | Responsibility                                                 | Trust boundary        |
|-----------------|----------------------------------------------------------------|-----------------------|
| Client          | Sends user message, timezone, and authenticated request.       | Untrusted input       |
| FastAPI gateway | Owns request validation, streaming, and service orchestration. | Application boundary  |
| Trend service   | Uses allow-listed SQL sources and computes summaries.          | Trusted calculation   |
| TimescaleDB     | Stores Django-created model output records.                    | Private data boundary |
| Ollama          | Turns trusted compact data into an explanation.                | No database access    |

## 2. Existing Data Architecture

The service reads existing Django/Timescale model-output tables. Each query is constrained by user\_id, date range, and a fixed metric-to-source mapping. Because source tables may have more than one record per date, the SQL selects the latest time value for each day.

| Chat metric | Existing table                  | Daily value      | Supporting fields                                                       |
|-------------|---------------------------------|------------------|-------------------------------------------------------------------------|
| Recovery    | ts_model_output_recovery_score  | recovery_score   | hrv_score, sleep_score, fatigue_score, interpretation_band, key_drivers |
| Readiness   | ts_model_output_readiness_index | readiness_score  | ri_base, ri_final, readiness_zone, penalty_triggers, key_drivers        |
| Injury risk | ts_model_output_risk_score      | injury_risk_prob | risk_band, risk_zone, model_version, key_drivers                        |

### Daily retrieval pattern

SELECT DISTINCT ON (date) ... WHERE user\_id = \$1 AND date BETWEEN \$2 AND \$3 ORDER BY date, time DESC.  
This retains one latest model output per athlete per local calendar day, while the outer query restores chronological order.

| Data rule      | Implementation                                                                                                                                      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity       | X-Athlete-ID is validated as a positive integer compatible with Django auth.User.                                                                   |
| Source control | Table and column identifiers originate only from the METRICS allow-list, not client text.                                                           |
| Null handling  | Null daily scores are excluded; availability is surfaced in data_quality.                                                                           |
| Risk semantics | A declining injury-risk probability is interpreted as improving. A 0-1 probability uses a 0.03 stability threshold; score-scale risk uses 3 points. |

### 3. Request and Analytics Flow

Retrieval completes before response streaming begins. This avoids partial answers based on unverified facts and keeps the LLM out of data-access decisions.



| Stage                      | Input                                 | Output                                          |
|----------------------------|---------------------------------------|-------------------------------------------------|
| Authorization              | X-Athlete-ID from trusted gateway     | Validated Django user_id                        |
| Intent and date resolution | Natural-language message and timezone | Allow-listed metric plus calendar range         |
| Trend query                | user_id, metric source, dates         | Chronological latest-daily values               |
| Summary                    | Daily values                          | Average, range, change, direction, completeness |
| Prompt assembly            | User question + analytics payload     | Bounded facts-only LLM context                  |
| Streaming                  | Ollama chat response                  | Concise non-medical explanation                 |

#### Example analytics contract

The chat model receives a JSON payload containing metric, label, period, daily\_values, summary, comparison when requested, data\_quality, and source\_table. It does not receive a SQL connection, unfiltered records, or authority to calculate missing data.

## 4. API and Deployment Design

The FastAPI application exposes an authenticated chat stream, separate current-day intelligence endpoints, and a read-only database health probe.

| Endpoint                         | Role                                                      | Key controls                                                                           |
|----------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------|
| POST /chat/                      | Historical trends and comparisons                         | Trusted identity header, fixed intent parser, parameterized queries, analytics context |
| POST /intelligence/{engine_type} | Current-day recovery/readiness/risk/overall model outputs | Allow-listed engine name; no score-history retrieval                                   |
| GET /health/db                   | Operational readiness                                     | SELECT 1 only; no credentials or health data returned                                  |

### Recommended deployment topology

| Layer         | Recommended implementation                                                                                              |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| Edge          | TLS terminator / API gateway validates JWT and injects X-Athlete-ID.                                                    |
| Application   | FastAPI container with pooled asyncpg connections and environment-provided DATABASE_URL.                                |
| Database      | PostgreSQL with TimescaleDB; TLS required; least-privilege read account for the chat service.                           |
| Model runtime | Ollama instances with separate chat, recovery, readiness, risk, and overall model mappings.                             |
| Observability | Request correlation, latency, token metrics, error class logging, and an audit store that avoids raw sensitive records. |

Network boundary: only the FastAPI service can reach the database; the Ollama runtime receives serialized, minimal context rather than database credentials or direct connectivity.

# 5. Security, Reliability, and Delivery

Health-adjacent analytics requires strong separation between user-controlled text, trusted calculations, and the model response.

| Control          | Required behavior                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Authorization    | Validate JWT upstream; bind X-Athlete-ID to the authenticated subject; never accept a user id from model output or arbitrary payload fields. |
| Database access  | Use TLS, a least-privilege role, parameterized values, and an allow-list for identifiers. Consider database RLS as defense in depth.         |
| LLM safety       | System message prohibits inventing dates or scores, medical diagnosis, raw database exposure, and missing-data calculation.                  |
| Privacy          | Do not place raw health records in logs or runtime training datasets. Store audit metadata and consent-aware retention controls.             |
| Failure handling | Return 503 for unavailable database, 422 for invalid timezone/message, and clear limited-data language when days are missing.                |
| Testing          | Cover numeric identity validation, date ranges, empty results, duplicate daily records, risk direction, and cross-user access attempts.      |

## Delivery sequence

| Phase         | Outcome                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------|
| 1. Foundation | Deploy connection configuration, health probe, Django-table queries, and deterministic trend tests.                    |
| 2. Experience | Add prior-period and metric comparisons, response observability, and compact conversation summaries.                   |
| 3. Enrichment | Introduce permission-filtered vector context only for approved notes and only when it improves a non-numerical answer. |

## Final design position

RhythmX should preserve the Django application as the system of record for model outputs. The intelligence API acts as a read-only, identity-scoped analytics and explanation layer: database and Python establish facts; Ollama communicates those facts safely.